MIL-E-1/1245B
3 December 1968
SUPERSEDING
MIL-E-1/1245A
2 May 1960

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6CB6A 1/

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Pentode, miniature, RF sharp cutoff

Outline	---	5-2 (EIA) Ⓑ
Base	---	E7-1
Envelope	---	T5-1/2
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7
Element	---	g1	k	h	h	a	g2	g3 and internal shield (see note 1)

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ec3	Ehk	Rk	Pp	Pg2	Ⓑ Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	v	Ohms	W	W	ft
Maximum:	6.9	330	---	165	---	100	---	2.3	0.55	(see
Minimum:	5.7	---	---	---	---	---	---	---	---	note 3)
TEST CONDITIONS:	6.3	125	0	125	0	---	56	---	---	---

GENERAL:

Qualification - Required

1/ See note 4

Ⓑ denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						MIN	MAX	
	Quality conformance inspection, part 1							
1231	Emission	Eb = Ec1 = Ec2 = Ec3 = 10 Vdc; Rk = 0 (see note 2)	0.65	II	Is	35	---	mAdc
1256	Electrode current (1) (anode)		0.65	II	Ib	8.0	18.0	mAdc
1266	Total grid current	Rg1 = 0.25 Meg; Eb = 200 Vdc; Ec2 = 150 Vdc; Rk = 56; Ec1 = -1.5 Vdc (see note 2)	0.65	II	Ic1	0	-1.0	μAdc
1306	Transconductance (1)		0.65	II	Sm	6,000	10,000	μmhos
1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1211	Insulation of electrodes		4.0	S3	R	10	---	Meg
1031	Low-frequency vibration	Eb = 200 Vdc; Ec2 = 150 Vdc; Rp = 2,000; Rk = 0; Ec1 = -2 Vdc	6.5	S3	Ep	---	250	mVac
1256	Electrode current (2) (anode)	Rk = 0; Ec1 = -6.5 Vdc	4.0	I	Ib	---	600	μAdc
1256	Electrode current (screen-grid)		6.5	S3	Ic2	---	6.0	mAdc
1301	Heater current		6.5	S3	If	275	325	mA
1306	Transconductance (2)	Ef = 5.7 Vdc	6.5	S3	Sm	5,200	---	μmhos
1531	Direct interelectrode capacitance	Shield No. 316	6.5	S3	Cg1-p	---	0.015	pF
					Cin	5.0	8.0	pF
					Cout	2.1	3.5	pF
1336	Heater-cathode leakage	Ehk = ±100 Vdc	6.5	S3	Ihk	---	20	μAdc
1121	Base strain		---	---	---	---	---	---
2126	Glass strain		4.0	I	---	---	---	---
1105	Permanence of marking		---	---	---	---	---	---
Ⓑ 1369	Heater warmup		6.5	S3	t	6.8	15	sec

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						MIN	MAX	
	Quality conformance inspection, part 3							
1501	Intermittent life	Group A; Ehk = 180 Vdc; Eb = 300 Vdc; Ec2 = 150 Vdc; Rk = 270	---	---	---	---	---	---
---	Life-test end point (intermittent) (500 hours)	Transconductance (1)	---	---	Sm	4, 800	---	μmhos

NOTES:

1. *Grid no. 3 has not been designed for control or gating purposes.*
2. This test to be performed at the conclusion of the holding period.
3. See "Reduced pressure (altitude) rating," and altitude, maximum peak voltage. Ⓑ
4. The tube described by this TSS is a substitute for tube type 6CB6, MIL-E-1/1245A.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - EL
Navy - SH
Air Force - 11, 17, 85
DSA - ES

User activities:
Army - MU
Navy - AS, OS, MC CG
Air Force - 19

Preparing activity:
Navy - EC

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